Data Warehouse Layers: Bronze, Silver & Gold

	Bronze Layer	Silver Layer	Gold Layer
Definition	**Raw, unprocessed data as-is from sources**	**Clean & standardized data**	**Business-Ready data**
Objective	**Traceability & Debugging**	**(Intermediate Layer) Prepare Data for Analysis**	**Provide data to be consumed for reporting & Analytics**
Object Type	**Tables**	**Tables**	**Views**
Load Method	**Full Load** (Truncate & Insert)	**Full Load** (Truncate & Insert)	**None**
Data Transformation	**None (as-is)**	• Data **Cleaning** • Data **Standardization** • Data **Normalization** • **Derived Columns** • Data **Enrichment**	• Data **Integration** • Data **Aggregation** • **Business Logic & Rules**
Data Modeling	**None (as-is)**	**None (as-is)**	• **Star Schema** • **Aggregated Objects** • **Flat Tables**
Target Audience	• **Data Engineers**	• **Data Analysts** • **Data Engineers**	• **Data Analysts** • **Business Users**

Development Workflow for Each Layer

Bronze Layer			
Analysing	**Coding**	**Validating**	**Docs & Version**
Interview source systems Experts	Data **Ingestion**	Data **Completeness & Schema** Checks	**Documenting** Versioning in **GIT**

Silver Layer			
Analysing	**Coding**	**Validating**	**Docs & Version**
Explore & **Understand** the Data	Data **Cleansing**	Data **Correctness** checks	**Documenting** Versioning in **GIT**

Documentation produced at this stage: Data Flow, Data Integration

Gold Layer			
Analysing	**Coding**	**Validating**	**Docs & Version**
Explore & **Understand** the Business Objects	Data **Integration**	Data **Integration** Checks	**Documenting** Versioning in **GIT**

Documentation produced at this stage: Data Model, Data Catalog, Data Flow

Source System Interview

Business Context & Ownership
Who **owns** the **data**?
What **Business Process** it supports?
System & Data **documentation**
Data **Model** & Data **Catalog**

Architecture & Technology Stack
How is data **stored**? (SQL Server, Oracle, AWS, Azure, …)
What are the **integration capabilities**? (API, Kafka, File Extract, Direct DB, ...)

Extract & Load
Incremental vs. **Full Loads**?
Data **Scope** & **Historical** Needs
What is the expected **size** of the **extracts**?
Are there any data **volume limitations**?
How to **avoid impacting** the source system's **performance**?
authentication and **authorization** (tokens, SSH keys, VPN, IP whitelisting, …)